# UNIT - 5 PROJECT IMPLEMENTATION AND REVIEW

**Introduction:**Project Implementation and Review is the most crucial stage in the life cycle of any project. While planning is about deciding what to do, implementation is about actually doing it.

Project Implementation refers to the process of executing the project plan and putting all planned activities into action. It involves allocation of resources such as manpower, materials, money and time, coordination of various departments, and effective leadership to achieve the project objectives. In the context of Human Resource Planning and Management, it means implementing HR projects like recruitment drives, training and development programs, performance appraisal systems, or implementation of HRIS.

The success of implementation depends on effective communication, proper monitoring, teamwork, and timely decision-making.

Project Review, on the other hand, is the systematic process of evaluating the progress and performance of the project. It is done during and after implementation to check whether the project is moving in the right direction, within budget and on time. Review helps to identify deviations, problems, and bottlenecks, and to take corrective measures.

In short, implementation converts ideas into reality, and review ensures that the reality matches the expected goals. Both are essential for successful completion of any project and for organizational growth.

Objectives of Project Implementation and Review:

1. To execute the project plan effectively and efficiently.
2. To ensure optimum utilization of human and material resources.
3. To monitor progress and control costs and time.
4. To identify problems and take corrective actions immediately.
5. To evaluate the overall success and learn lessons for future projects.

This is why Project Implementation and Review is considered the backbone of project management.

## FORMS OF PROJECT ORGANISATION

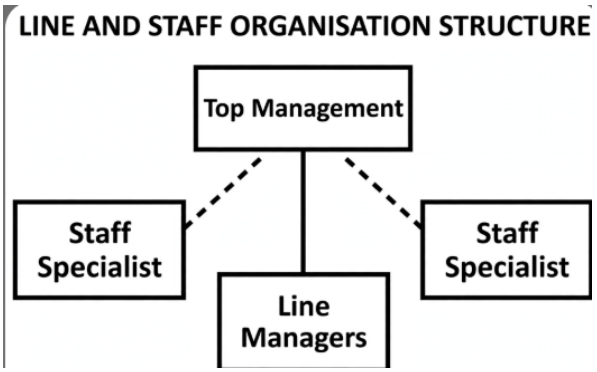
Every project is done by people. For a project to be successful, we need to decide who will do what work, who will report to whom, and who has the power to take decisions. This arrangement of people is called Project Organisation.

If the people are not organised properly, work gets delayed, confusion happens, and the project may fail. So, to avoid this, different forms of organisation structures are used based on the size and type of the project.

The main 5 forms are:

**1. Line and Staff Organisation:** Line people are the main decision makers. Staff people are experts who give advice to line people. Staff people cannot give direct orders.

- ❖ Simple and clear.
- ❖ Experts are always available for help.




**Advantages:**

Expert advice from staff is available

Line managers can focus on main work

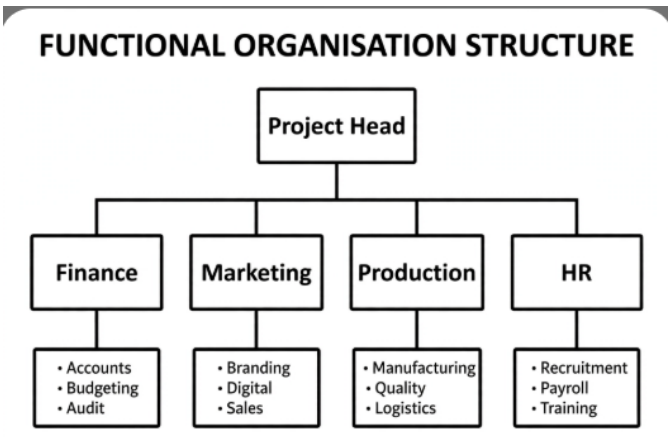
**Disadvantages:**

Conflict may arise between line and staff

Staff advice is not compulsory, so may be ignored

**2. Functional Organisation: Here,** the company is divided based on the type of work. Like one department for design, one for production, one for testing. People from each department work together for the project.

- ❖ Best when special skills are needed.
- ❖ Work is done with expert quality




**Advantages:**

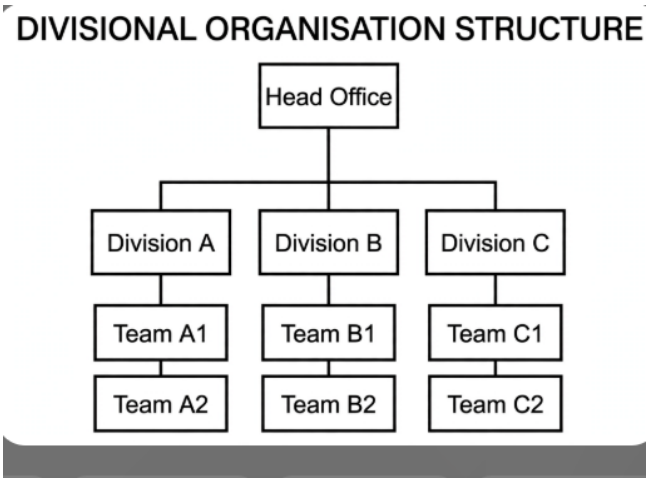
- ❖ Work is done by specialists — high quality
- ❖ Easy to manage and control

**Disadvantages:**

- ❖ Slow decision making
- ❖ No single person responsible for full project

**3. Divisional Organisation**: Here, the company is divided based on product or place. Each division has its own team with all facilities. Each division works like a small independent company.

- ❖ Each division focuses only on its own product.
- ❖ Easy to manage big companies.




**Advantages:**

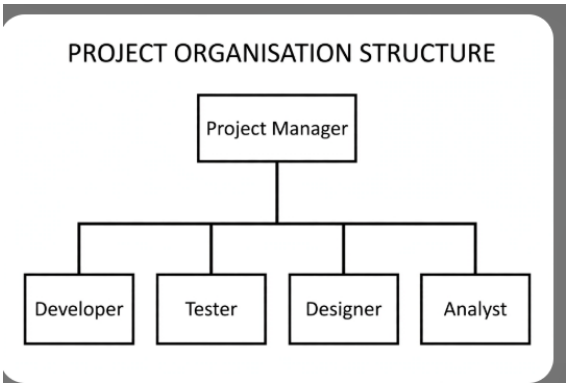
- ❖ Full focus on one product or division
- ❖ Quick response to market changes

**Disadvantages:**

- ❖ Costly — each division needs separate staff
- ❖ Duplication of resources

**4. Project Organisation (Pure Project):** Here, a separate team is made only for one project. One Project Manager is the full boss. He has a dedicated team of members who work only for that project. When the project is over, the team is also closed.

- ❖ Full control to Project Manager.
- ❖ Project gets completed very fast.




**Advantages:**

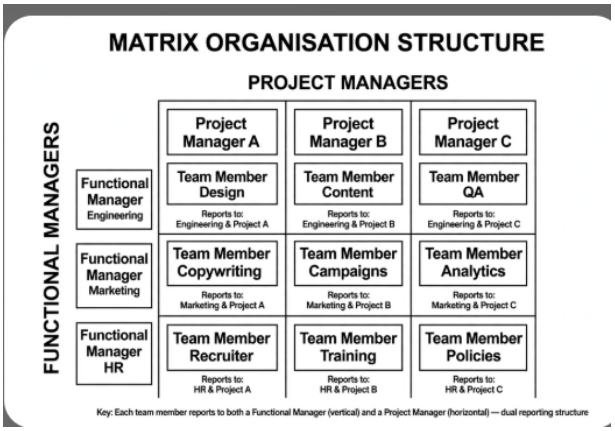
- ❖ Full authority to Project Manager
- ❖ Fast completion of project

**Disadvantages:**

- ❖ Costly — dedicated team for one project
- ❖ Team becomes jobless after project ends

**5. Matrix Organisation**: This is a mix of Functional and Project structure. In this, one person has two bosses. One boss is from his main department and second boss is the Project Manager. So he has to report to both.

- ❖ Best use of people and resources.
- ❖ Sometimes confusion happens because of two bosses.




**Advantages:**

- ❖ Best use of skilled people
- ❖ Flexible and balanced structure

**Disadvantaes:**

- ❖ Dual reporting — one person has two bosses
- ❖ Confusion and conflict between managers

## PROJECT PLANNING:

**Definition:P**lanning is an intellectual process, the conscious determination of courses of action, the basing of decisions on purpose, facts, and considered estimates.
Meaning.Planning means predicting or forecasting the future activities and jobs in advance.

Project Planning refers to determining various tasks and activities to be performed on the project and scheduling them in order by assigning the respective time for each and every activity.

It refers to defining and describing the various activities and the end products of a project to achieve the overall objective.

- Under Project Planning, the following need to be defined:
- The goals of the project and respective activities required to accomplish them.
- Documentation of various estimates for the purpose of controlling and correcting.
- Documentation of the agreements by the affected groups and individuals.

**FUNCTIONS OF PROJECT PLANNING:**

- Various activities are Organising and assigning their responsibilities.
- Serves as a platform for coordination and communication among the individuals.
- People realises the importance of timely completion of schedule.
- Control and correction is established.

**OBJECTIVES OF PLANNING :**

- Various complexities involved in the project are arranged in a series of activities to overcome them
- Renders an estimation of time and resources required in each activity
- Making adequate provision for Risk and uncertain events
- Effective resource utilisation
- Ensuring availability of good quality data to assist decision making
- To ensure that activities are in line with the overall objectives

**PROCESS OF PROJECT PLANNING** - Project Planning is the first and most important step in Project Management. Before starting any project, we have to think and decide in advance what to do, how to do, when to do and who will do it.
According to your notes, planning is an intellectual process - the conscious determination of courses of action, basing decisions on purpose, facts and considered estimates.

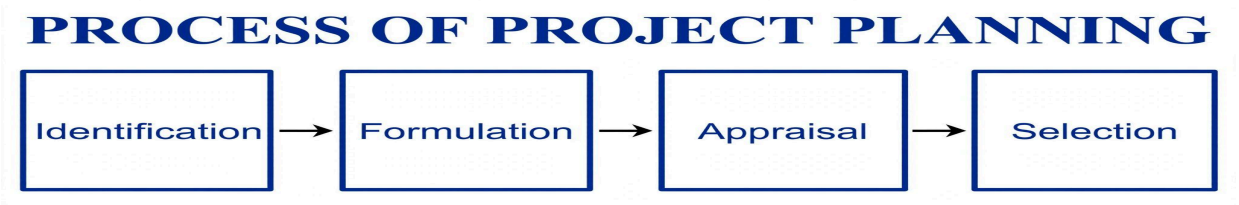
The Process of Project Planning explains the systematic steps followed to convert an idea into a feasible project. There are 4 main stages.

**1. Identification** : It is the first stage where the project idea is identified. The need or opportunity is recognised.

**2. Formulation** : The identified idea is formulated in detail. Objectives, resources, cost and time are clearly defined and a project proposal is prepared.

**3. Appraisal :** The project is evaluated from all angles - technical, financial, market and economic feasibility. It checks whether the project is viable or not.

**4. Selection :** After appraisal, the most suitable and viable project is finally selected for implementation.




**TECHNIQUES OF PROJECT PLANNING:**

**1. Multi-level Scheduling:**Means making schedules at different levels. As per your notes 3 levels:

- ★ **Master Project Schedule** - for top management
- ★ **Functional Area Schedule** - for departments
- ★ **Schedule Work Package -** detailed schedule for small works

**2. Multiple Project Scheduling:** When a company does many projects at same time, resources (money, people, machines) must be shared properly among all projects.

**3. Material Requirement Planning (MRP**): Used to estimate what materials are needed. It uses inventory data, material bill, and master production schedule.

**4. CATS and RATS:**Two ways to schedule time:

- ★ **CATS** - Committed Activity Target Schedule
- ★ **RATS** - Reserved Activity Target Schedule
  Both need to be revised if cost changes.

**5. Network Analysis :**Activities and events are drawn in a diagram to show planning and scheduling. Makes control easy.

PERT and CPM are the main techniques - lifeline for managers for turnkey projects, helps to meet deadlines on time.

**6. Allocation of Resources:** Means distributing available resources (men, material, money, machines) to different activities properly.

**7. Resource Levelling:** Means balancing resources so that no resource is over-used or idle. To make work smooth.

**8. Work Packages:**Big project is broken into small small sets of tasks. It may be 2-level, 3-level breakdown.
Each work package has: Clear SOW (Statement of Work), Resource, Time, Cost, Quality, Responsibility, Outcomes, Risk, Inputs.

**9. Bar Chart (Gantt Chart) and Line of Balance:**Bar Chart: Also called Gantt chart or multiple activity chart. Oldest method used for formal planning. Shows which activity starts when and ends when.

**Line of Balance:** Used when same work is repeated many times.

**10. Linear Programming :**A mathematical technique to find best use of resources and time.

No single technique is enough. A good project manager should select and combine suitable techniques according to the nature of the project. Proper use of these techniques ensures that the project is completed on time, within budget, and with good quality.

**Techniques make the project planning systematic, easy to control and successful.**

## PROJECT CONTROL

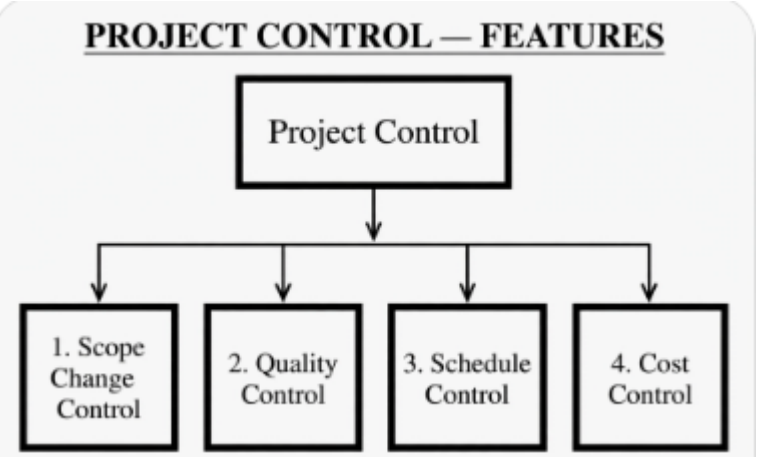
Project Control is the most difficult and central job of a project manager.It means checking and guiding the project from starting to ending to make sure everything goes as per plan.

**Definition:**“ Project Control is the process that ensures project objectives are met — work is done on time, within cost, and with good quality. “

**FEATURES OF PROJECT CONTROL :**

**1. Scope Change Control** :Controlling extra work. Don't allow unnecessary changes in project.

**2. Quality Control :** Checking quality. Work should be as per standard and good quality.




**3. Schedule Control :**Controlling time. Work should finish on time as per plan.

**4. Cost Control :** Controlling money. Project should finish within budget.

**OBJECTIVES :** There are three different types of objectives

**1. Primary Objectives:**

(1) Cost Control
(2) Schedule Control
**(3) Quality Control**

**2. Secondary Objectives**:

(1) Risk Control
(2) Communication Control
(3) Resource Control

**3. Tertiary Objectives:**

(1) Scope Control

**Types of Project Control:**There are 3 main types:

**1. Cybernetic Control:** Works like a thermostat, with feedback correcting the project automatically while it runs.

- Uses feedback to monitor the project.
- Makes continuous adjustments based on project output.
- Helps maintain stability and achieve goals.

**2. Go / No-Go Control:**A checkpoint test at the end of a stage. The project continues only if it passes.

- Checks whether the project meets time and cost standards.
- If standards are met → Go
- If standards are not met → No-Go / corrective action

**3. Post Control:**done after completion, to learn lessons for future projects.

- Applied after the project is completed.
- Reviews the completed project and its results.
- Also called Post-Mortem Control.

Project control is a continuous process of measuring performance, comparing it with the plan and taking corrective action. Effective control, supported by timely and reliable information, helps a project achieve its objectives within the planned time, cost and quality.

**REASONS FOR INEFFECTIVE CONTROL IN PROJECTS :It** can be seen that effective control is crucial in achieving the objectives of the project.These controls differs in different situations.The three probable reasons are :

1. **Project characteristics** : projects are unique, temporary and uncertain, so standard controls may not fit.
2. **People problem** : resistance, lack of skills, poor commitment or conflict among team members.
3. Poor control and information system
    a. Delayed Reporting Performance: problems are known too late to correct.
    b. Inappropriate Detail level : too much or too little information confuses managers.
    c. Unreliable set of information: wrong data leads to wrong decisions.

## HUMAN ASPECTS OF PROJECT MANAGEMENT

The success of a project depends not only on money, technology and materials, but also on people.Human aspects of project management deal with:

- Managing people
- Motivation
- Leadership
- Authority
- Teamwork
- Communication
- Commitment
- Group functioning

A project manager should maintain good relationships with the team to achieve project objectives.Human aspects deal with **managing people effectively to achieve project objectives**.

**1. Project Authority**

- Authority is the **power to control and guide project activities**.
- The project manager uses authority to **assign work and make decisions**.
- Authority should match the manager's responsibility.

**2. Project Orientation**

- Focuses on **directing team members towards project goals**.
- Success depends on **good leadership and teamwork**.
- The project manager should build good relationships with team members.

**3. Commitment & Motivation**

- **Motivation** encourages employees to work effectively.
- **Commitment** means dedication towards project goals.
- Both are important for successful project completion.

**4. Group Functioning**

People from different departments may work together in a project.

**Types:**

- **Formal Group** – formed according to organisational rules.
- **Informal Group** – formed voluntarily by people.

**5. Types of Project Groups**

- **Horizontal Group** – same level, different departments.
- **Vertical Group** – same department, different levels.
- **Mixed Group** – combination of horizontal and vertical groups.

**6. Building Effective Groups**

An effective group requires:

- Mutual trust
- Open communication
- Cooperation
- Support
- Conflict resolution
- Taking initiative

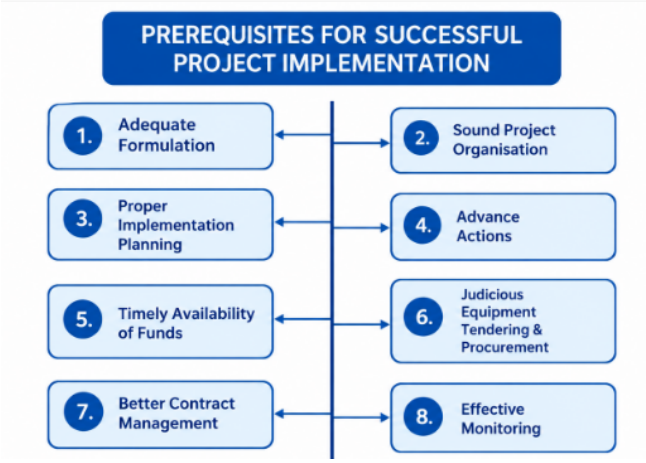
**Conclusion:**Good leadership, motivation, commitment and teamwork help people work together and achieve project objectives successfully.

**Prerequisites for Successful Project Implementation:** "Project implementation is the process of executing the planned project activities. For successful implementation, certain prerequisites must be fulfilled to avoid failure, delays and cost overruns."

**1. Adequate Formulation:** Project should be properly formulated. Fails due to lack of experience, poor decision, over-estimation of benefits and under-estimation of cost.

**2. Sound Project Organisation:** Need competent leader, equal distribution of Authority & Responsibility, and clearly defined systems.

**3. Proper Implementation Planning** : Create a perfect plan and implement according to the plan.




**4. Advance Actions** : Know latest technology trends, identify technology collaborators/consultants, arrange infrastructure facilities

**5. Timely Availability of Funds :** Funds must be available at the right time without delay.

**6. Judicious Equipment Tendering & Procurement** : Maintain balance between Foreign and Indigenous suppliers. Check competence of contractor before giving contract.

**7. Better Contract Management** : Maintain discipline. Authorities must have power to control. Transfer contract to third party if delay is anticipated.

**8. Effective Monitoring** : Continuous monitoring of project work to avoid delays and mistakes.

## PROJECT REVIEW

A **project review** is a systematic examination of a project at different stages of its life cycle.It helps the project manager to:

- Check the progress of the project.
- Identify problems at an early stage.

- Check whether the project is meeting its objectives.
- Take corrective actions when required.
- Improve the chances of successful project completion.

Project reviews provide an opportunity for the project team to discuss the **design, performance, progress, problems and improvements** of the project.

**Types of Project Reviews:** A project manager generally conducts **three types of project reviews**:

1. Status Review
2. Design Review
3. Process Review

**1. Status Review :** A **status review** checks the current progress of the project.It helps the project manager to know whether the project is progressing according to the **planned time, cost and scope**.

**Types of Status Review**

**a) Cursory Review**

- A quick and brief review of the project.
- Used to check the general status of the project.

**b) Comprehensive Review**

- A detailed examination of the project.
- Checks important areas such as **cost, time, performance and scope**.

**Important Points:**

- Status reviews should be conducted at regular intervals.
- Reviews should also be conducted whenever major problems occur.
- The project manager should avoid conducting unnecessary reviews.
- Reports should contain meaningful numerical information.
- The reporting system should identify deviations from the planned limits.

**2. Design Review :** A **design review** is conducted to check whether the design of the product or service meets the required performance standards.It is usually conducted at **major milestones** of the project.

**Main Objectives:**

- To check the quality of the product design.
- To ensure that the product can achieve the required performance.

- To identify design problems at an early stage.
- To improve the overall design.

**A Design Review Covers:**

- ❖ **Manufacturing and Servicing**
  - ➢ Checks whether the product can be manufactured and serviced easily and economically.
- ❖ **Safety**
  - ➢ Checks the safety of operators and maintenance personnel.
  - ➢ Helps to reduce possible damages and risks.
- ❖ **Scope and Technical Specifications**
  - ➢ Checks whether the design meets the required scope and technical specifications.

**Process Review:**A **process review** checks whether the project processes are being carried out as planned.

**Main Objectives**

- To check whether processes are working properly.
- To identify problems in the process.
- To find opportunities for improvement.
- To ensure that project activities follow the planned procedures.

A process review is generally conducted **periodically or at major milestones** of the project.

**Advantages of Project Review:**Project reviews help the project manager to monitor and improve the project during implementation.

**Main Advantages:**

1. **Achievement of Business Goals:**Checks whether the project can achieve its planned business goals.
2. **Proper Implementation of Rules:**Ensures that organisational rules are properly understood and followed.
3. **Checking Project Worth:**Helps to determine whether continuing the project is worthwhile before entering into major commitments.
4. **Effective Project Management:**Checks whether the project is being managed effectively.
5. **Early Problem Identification:**Helps to identify and solve problems before they become serious.
6. **Improved Project Performance:**Helps improve the way the project is planned and managed.

7. **Better Team Performance:**Ensures that team members understand their responsibilities and follow project guidelines.

**Conclusion:**Project reviews help the project manager identify problems early, monitor progress and take corrective actions for successful project completion.

**Performance Evaluation of Projects:**

A. Control of In-Progress Projects :

- - This means checking the project while it is still going on.
- - Even with good planning, problems can happen like delays, extra cost, or poor quality.
- - So we must regularly review the project.

By having an Internal Control System with 5 parts:

1. Control Environment - company culture and rules
2. Risk Assessment - finding what can go wrong
3. Control Activitie s- actions to prevent problems
4. Information & Communication - sharing info properly
5. Monitoring - checking if everything is working

**B. Post Completion Audit / PCA:** Also called "post-project evaluation" or "post-completion audit".

- **Project Appraisal** = checking estimates before the project. It's for the future.
- **Project Evaluation** = checking results after the project. It's for the past.
- It compares planned vs actual cost, time, resources and benefits.
- Helps the organization learn and improve for next time.

**C. Abandonment Analysis :**

- - Done after project starts to check if it should be stopped.
- - If the project is not making profit, or future cash flow will be negative, we should close it early.
- - Decision depends on: political changes, tax, economy, and other country factors.

# Abandonment Analysis

**Meaning:**Abandonment analysis is the process of deciding whether a project should be **continued or stopped before completion**. It is done during project implementation, after the project has started. When a project is found to be less profitable than expected, the decision to abandon or terminate it early is considered.

**Rule for Abandonment:** A project should be abandoned only when its **net discounted expected future cash flow becomes negative**, or is less than its salvage value.

- If the present value of future cash flows is **less than the salvage value → Abandon**
- If the present value of future cash flows is **greater than the salvage value → Continue**

### When a Project May Be Abandoned

1. **Loss of strategic importance:** the project has lost its long-term and short-term strategies, so it no longer fits the organisation's goals.
2. **Scarcity of resources:** the required resources become scarce and restricted.
3. **Lag behind schedule:** the progress of project implementation continuously lags behind the schedule.
4. **Infeasibility due to changed circumstances:** completing the project has become unfeasible due to changed circumstances.

### Factors Affecting the Decision

- Political changes
- Tax
- Economic conditions
- Other country-specific factors

**Example:** A project can be sold now for ₹40 lakh (salvage value), but its future cash flows are worth only ₹30 lakh today. Since continuing gives less than selling, the project should be **abandoned**.

Abandonment is not a failure but a **rational managerial decision to avoid further losses**. It helps in saving resources, minimising risk, and reallocating funds to more profitable opportunities.

### Causes of Project Failure and How Successful Implementation Can Be Ensured

**Introduction:** A project is said to fail when it is not completed on time, goes beyond the budget, does not have the required quality, or does not achieve its objectives. Most project failures can be avoided if we know the causes and take proper steps during implementation.

## Causes of Project Failure

1. **Poor planning:** The objectives are not clear, and the time and cost estimates are wrong.
2. **Unclear or changing scope:** If requirements keep changing, the time and cost keep increasing.
3. **Shortage of resources:** There is not enough money, skilled people, materials or equipment.

4. **Weak project management:** The project manager lacks leadership, coordination and decision-making skills.
5. **Poor communication:** The team, client and stakeholders do not understand each other, which leads to mistakes and delays.
6. **Lack of support from top management:** Without their commitment and funds, the project cannot move forward.
7. **Ineffective control:** Problems are found too late because progress is not checked regularly.
8. **Poor risk management:** Risks are not identified in advance, so unexpected problems stop the work.
9. **Wrong choice of technology:** The technology is unsuitable, or the team does not know how to use it.
10. **Team problems:** Conflicts, low motivation and lack of skills reduce performance.
11. **Poor quality:** Errors and defects make the customer unhappy.
12. **Changes in outside conditions:** Changes in politics, economy, tax or government policy can make the project unprofitable.

# How Successful Implementation Can Be Ensured

1. **Make a proper plan:** Fix clear objectives, scope, time and budget, and use tools like PERT and CPM.
2. **Do a proper appraisal:** Check technical, financial, market and social feasibility before starting.
3. **Choose a suitable organisation structure:** Select the functional, project or matrix structure that fits the project.
4. **Select the right team:** Give the right task to the right person, and define roles and responsibilities clearly.
5. **Get top management support:** Obtain their commitment, authority and funds.
6. **Control changes:** Allow a change only after studying its effect and getting approval.
7. **Maintain good communication:** Hold regular meetings and share progress reports with everyone concerned.
8. **Monitor and control regularly:** Compare actual progress with the plan and take corrective action quickly.
9. **Manage risks:** Find risks early, prepare backup plans and watch them throughout the project.
10. **Motivate the team:** Encourage teamwork and solve conflicts quickly.
11. **Conduct project reviews:** Status, design and process reviews help find problems early.
12. **Use abandonment analysis:** Stop a project that has become unprofitable instead of wasting more money.
13. **Do a post-completion audit:** Check the results after completion and learn lessons for future projects.

# Conclusion

Project failure usually happens because of poor planning, weak control, unclear scope, shortage of resources and poor communication. A project can be successful if it has good planning, a strong team, effective communication, regular monitoring and timely review. When these steps are followed, the project is completed on time, within budget and with the required quality.

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